

Nanyang Technological University

NTU Entrepreneurship Academy

Kurtosis Labs Business Proposal

Company	Kurtosis Labs
Current stage	Pre-incorporation, pre-revenue deep-tech startup at the proof-of-concept ¹ stage. We are building risk-managed DeFi ² strategies and decision infrastructure ³ using hybrid quantum-classical ⁴ decision infrastructure for on-chain capital allocation ⁵ .

1. A clear description of your problem statement

Investing in Decentralised Finance (DeFi) faces a huge problem, too much risk. Investors are at risk of losing their entire capital to forces such as impermanent/liquidity loss⁷, tail-events⁸, and high fees.

Looking at yield farming⁶ in particular, over 70% of yield positions placed are not adequately rewarded for the risk they bear. Simply put, yield farmers are highly likely to lose the capital they invest into funds. Existing institutions only take action to recuperate losses, but not enough is done yet to prevent them in the first place.

The main reason why this exists is due to a lack of governance infrastructure⁹ and fragmented decision-making. Kurtosis Labs is addressing this issue by building actual governance and control tools, deploying risk-management software for institutions.

2. A detailed description of your team's proposed solution

Kurtosis Labs is a deep-tech startup focusing on risk management-as-a-service¹⁰ solutions. We are building the risk management frameworks, control planes¹¹, and strategies to manage risk in DeFi. This all translates into our main product offering, the Decision Engine¹².

The Decision Engine comprises of 3 layers:

1. Market Intelligence¹³ - Gathers market data, on-chain data, and protocol data¹⁴.
2. Risk Policy¹⁵ - Apply safety checks including leverage limits¹⁶, liquidity limits, fund exposure¹⁷.
3. Portfolio Optimisation¹⁸ - Using quantum-classical orchestration¹⁹, we compare portfolio combinations to identify the best options to achieve sustainable yields.

These are products that we plan to develop in-house and through partnerships with existing protocols and chains. By and large, we will be an ecosystem of products designed to transform institutional risk management in decentralised finance.

3. Identify your competition (companies, products, technologies or status quo)

Competitor / Alternative	What they do	Where are they strong	Where are we different
Gauntlet	Advises crypto protocols and vault ²⁰ ecosystems on risk.	Strong reputation in DeFi and deep market knowledge	Our Decision Engine has multiple layers and not just research.
Steakhouse Financial	Builds vaults and risk frameworks for DeFi capital allocation	Clear governance, monitoring, and institutional-style reporting	We focus on the automation of risk management and create tools that replace existing manual workflows.
Sentora	Institution offering non-custodial ²¹ strategies, integrated risk-management, and research.	Strong at institutional DeFi access, risk analytics, and custom vault products across many protocols and chains.	Our classical-quantum infrastructure and quantum orchestration is better suited to curate strategies for portfolio optimisation.

4. If possible, present your POC or MVP²²

Website: kurtosis-labs.com

MVP: kurtosis-labs.com/mvp

5. A 100-word description of your project that can be shared publicly

Kurtosis Labs is a deep-tech startup using quantum orchestration to build risk management infrastructure in decentralized finance. Our goal is to make decentralized finance safer for money to be deployed by institutional and individual investors.

6. A description of team members' roles and responsibilities

Kai Ze Tam — Co-Founder / CEO: Background in CS, Quant Finance, and Quantum Research.

Leads company vision, strategy direction, fundraising, and quantum research. Kai owns the hybrid quantum-classical optimization architecture, and bridging traditional quant finance into DeFi.

Bryan Goh — Co-Founder / CPO: Background in CS and DeFi.

Owns platform engineering, including integrations, data ingestion, execution logic, and the interfaces between the quantum, DeFi, and execution layers.

Christen Cheng — Co-Founder / CTO: Background in CS and DeFi.

Leads the engineering team and heads DeFi research at Kurtosis Labs. Christen owns technical architecture and directs the build-out of vault strategies and risk-management systems.

Ryan Mansor Oh — Founding Engineer: Background in CS and Quant Finance.

Focuses on DeFi quantitative research and risk modelling. Ryan works on strategy design, simulation, and the research needed for vault curation.

Michelle Hanjaya — Data Engineer: Background in CS and Banking Tech.

Builds data pipelines²⁴, analytics, and visualization tools for the platform. Michelle supports internal and external product surfaces together with the engineering team.

7. One-year project timeline with clearly indicated milestones

Phase	Timeline	Milestones
Phase 1: Validation and product specification	Months 1–2	1. Finalize target user workflow and pilot requirements 2. Build one 10–12% target vault strategy sleeve 3. Stand up Solana²³ data pipelines and protocol adapters²⁵
Phase 2: Core MVP build	Months 3–5	1. Build the Decision Infrastructure MVP 2. Build the user interface to view vaults and decisions 3. Support 3 strategy sleeves in dry-run²⁶ and simulation mode²⁷
Phase 3: Integration and guarded testing	Months 6–8	1. Integrate with Ranger Labs and one additional execution environment 2. Run guarded testing on small-capital or test workflows 3. Secure 2–3 pilot partners
Phase 4: Pilot preparation	Months 9–10	1. Run a pilot for on-chain asset allocation support 2. Build internal risk reporting and decision records³² 3. Tighten strategy parameters for net yield and guardrail compliance³³
Phase 5: Pilot execution and iteration	Months 11–12	1. Complete at least one successful pilot 2. Expand to 5–10 strategy sleeves across multiple venues or wrappers 3. Target first \$1M–\$5M in pilot capital / TVL²⁸ influenced through deployments

8. Provide a detailed breakdown for the grant usage (up to S\$5,000)

The proposed grant budget is intentionally lean and focused on software, data access, and operating tools that shorten the path to a pilotable product. It is not intended for broad hiring. Where grant rules do not permit incorporation or administrative costs, that allocation can be redirected to cloud compute³¹ for prototyping and testing.

Item	Purpose	Amount (\$)	Notes
AI software tools (ChatGPT, Claude, Cursor)	Support research, drafting, coding assistance, and faster iteration during the POC and MVP build.	1,300	Directly improves engineering and strategy development speed.
Data, RPC ²⁹ , and API ³⁰ services	Reliable access to market, on-chain, and protocol state for testing, simulation, and live integration work.	1,400	Improves data quality and execution realism.
Operational subscriptions	Essential operating software such as domain, email, collaboration, monitoring, design, and documentation tools.	800	Keeps the team operational during the build phase.
Incorporation administrative costs / OR cloud compute	Either incorporation and required administrative setup, or redirected to cloud compute for prototyping, storage, and deployment if incorporation is not claimable.	1,500	Either-or line item depending on grant restrictions.
Total		5,000	

Annex

Technical Terminology

Superscript numbers in the proposal refer to the glossary below.

No.	Term	Plain-language explanation
1	Proof-of-concept (POC)	An early version built to test whether the core idea works. It is not yet a finished product.
2	DeFi	Short for decentralised finance: financial services built on blockchains without relying on traditional banks or brokers.
3	Decision infrastructure	Software and control systems that help decide where money should go and what rules must be followed.
4	Hybrid quantum-classical	Using both quantum computing and conventional computing together so each handles the part it is best at.
5	On-chain capital allocation	Deciding how funds should be placed across blockchain-based applications and strategies.
6	Yield farming	Moving crypto assets between protocols or strategies to earn returns such as fees, incentives, or lending yield.
7	Impermanent / liquidity loss	Losses that can happen when assets placed in liquidity strategies change in price or cannot be exited efficiently.
8	Tail events	Rare but severe market shocks that can cause unusually large losses.
9	Governance infrastructure	The rules, approvals, and control systems used to govern how financial decisions are made.
10	Risk management-as-a-service	Providing risk-management tools and systems as a service to other organisations.

11	Control plane	The central layer that sets rules, coordinates actions, and makes sure the wider system behaves correctly.
12	Decision Engine	Kurtosis Labs' core software system for gathering data, checking risk, and preparing portfolio decisions.
13	Market Intelligence	The data layer that collects market and protocol information needed for decision-making.
14	Protocol data	Information about the current state of a blockchain application, such as rates, liquidity, balances, or configuration.
15	Risk Policy	The set of safety rules and limits that the system must follow before capital can be deployed.
16	Leverage limits	Caps on how much borrowed money or leverage can be used in a strategy.
17	Fund exposure	How much of a portfolio is at risk to a single asset, venue, counterparty, or strategy.
18	Portfolio optimisation	Choosing the mix of positions that best meets a return goal while staying within risk constraints.
19	Quantum-classical orchestration	Coordinating quantum and classical computation so they work together in one decision workflow.
20	Vault	A pooled product that automatically deploys user funds according to a defined strategy.
21	Non-custodial	A setup where the user or institution keeps control of its own assets instead of handing custody to a third party.
22	Minimum viable product (MVP)	The smallest usable version of a product that can be tested with real users or partners.

23	Solana	A blockchain network commonly used for fast, low-cost on-chain applications.
24	Data pipeline	A system that gathers, cleans, moves, and prepares data so it can be used by software.
25	Protocol adapter	A connector that lets one system interact with a specific blockchain protocol.
26	Dry-run	A test execution that shows what would happen without moving real funds.
27	Simulation mode	A testing mode that models expected results before anything is deployed live.
28	TVL (total value locked)	The total dollar value of assets currently deposited in a protocol, vault, or strategy.
29	RPC (remote procedure call)	A service that lets software read blockchain data and send requests to a network.
30	API (application programming interface)	A standard way for one software system to communicate with another.
31	Cloud compute	Rented computing resources such as servers, storage, and processing power used for building and testing software.
32	Decision records	Saved logs that explain what decision was made, when it was made, and why.
33	Guardrail compliance	Staying within preset safety limits that stop the system from taking unsafe actions.